

An Anti-Collision protocol of RFID Based on Divide and Conquer Algorithm

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Abstract

Radio frequency identification (RFID) system is a wireless automatic identification mechanism. Now, it is widely applied in supply chain management, supermarket checkout process, public transportation, animal identification, the localization and tracking of objects and people, and toll ways. The tag collisions in RFID system is an unavoidable problem. The number of collision in previous tag anti-collision protocol algorithms is direct proportion to the number of tags. In this paper, we propose a new anti-collision protocol algorithm based on divide and conquer for RFID system. Our algorithm divides the successive collision bits into several two bits groups, use these groups to search which the collision group is and where the collision occurring, and identify all tags finally. From the experiment results, the number of tag collision in our algorithm only depends on the length of tags, and is independent of the number of tags. So, the time complexity our algorithm is $\text{BigO}(1)$.

Key Words: RFID, Anti-Collision, divide and conquer.

1 Introduction

Omnipresent tagging is a pattern that every object has its own unique tag. If many objects in the world can be uniquely and reliably identified by electronic tags at the same time, it would can track and identify any object to make omnipresent identification to create tremendous benefits. In recent times, omnipresent identification systems have become very common in access control and security applications fields. Radio frequency identification (RFID) system is a wireless automatic identification mechanism. RFID was developed very early and was widely used in many areas recently. RFID is firstly applied in the distribution industry, and then is applied to other areas such as supply chain management, supermarket checkout process, public transportation, animal identification, toll ways, and the localization and tracking of objects and people[1][2][3].

A RFID system consists of three main components as shown in Figure 1 [4][5]:

- Radio frequency (RF) tags, also known as transponders (transmitter/responder). A tag has a unique identification number (ID) that can be identified by networked electromagnetic readers, and is attached to the object that needs to be identified or to be counted. There are active tags and passive tags. Active tags have storage capabilities and are partly or fully provided with power sources. Active tags have the capability to communicate with other tags, and can initiate a dialogue of their own with the tag reader. On the other hand, Passive tags do not need any internal power source but are powered up by the radio waves emitted by the tag readers during the reading process. Tags consist mainly of a microchip to store the electronic codes of object and a coiled antenna to transmit the electronic codes of object with readers.

- A reader or transceiver (transmitter/receiver) consists mainly of an RF module to make active the tags, and a control unit to makeup the communication sequence with the tags and to transfer data between tags and the RFID middleware.

- A RFID middleware, which can makeup the communication sequence with the reads and to transfer data between reads and application systems. Middleware is an application or database, depending on the applications.

During the reading process in an RFID system, the reader broadcasts a request message to tags to ask for the unique tag ID or the detailed data saved in tags. Several tags send their response back to the reader after tags receive the request message. The reader must be able to identify tags as soon as possible. However, if two or more tags response simultaneously, messages transmitted simultaneously by readers or tags will collide such that the reader can't correctly receive the right messages because they communicate over a shared wireless channel. Therefore, an effective RFID must reduce these collisions and identify tags regardless of the occurrence of collisions by using anti-collision protocols.

There are reader collisions and tag collisions in the RFID system to reduce the identification rate (number of identification tags per time unit) [3].

Reader collisions take place when several neighbor readers try to interrogate the same tag at the same time, such that the neighbor readers interfere with one another, and the tag will not be able to respond to any readers. [6][7]. Reader collisions always can be reduced or avoided because the multiple neighbor readers can coordinate with each other to keep their distance long enough in advance. Tag collisions arise when several tags try to response their IDs to a reader simultaneously to prevent the reader to recognize any tag. However, low-functional passive tags lack power supply and communication ability. They only transmit their IDs to the reader by reflecting the waves emitted from the reader. They are unable to figure out neighbor tags and cannot detect collisions. Therefore, the tag collisions of passive tags cannot be resolved easily with methods based on Carrier Sense Multiple Access (CSMA) [8], and tag anti-collision protocols for passive tags are regarded as a key of the universal acceptance of the RFID system [7][9].

Tag anti-collision protocols can be divided into two kinds, ALOHA based anti-collision protocols and tree based anti-collision protocols. ALOHA based protocols decompose the tags transmit into distinct time slots to reduce the occurrence rate of tag collisions [11]. However, ALOHA based protocols are unable to reduce the tag collisions completely. On the other hand, the ALOHA based protocols have a problem that some specific tags may not be identified for a long time and produce the tag starvation problem. The tag anti-collision protocols based on tree use the decomposition mechanism to identify tags. They divide collision tags into two

non-overlapped subsets while tag collision occurs. Then, the reader identifies the subsets one by one. Repeating this kind decomposition until the reader receives tag IDs that have no any collisions [11]. Although the tag anti-collision protocols based on tree don't cause tag starvation problem, they have longer identification delay caused by the dividing procedure.

The divide and conquer algorithm is a popular algorithm, and is widely used in computer graphs, image and video segmentation, and linear algebra [13]. The divide and conquer algorithm divides a big problem of size n uniformly at into two small sub-problems of size k and $n-k$ that can be solved easily and quickly, repeating to the same solving steps to solve these small sub-problems, and then somehow combine their solutions to obtain a solution to the original problem [14]. We use the characteristic of the divide and conquer algorithm to propose a new anti-collision protocol algorithm based on divide and conquer for RFID system. Our algorithm divides the successive collision bits into several two bits groups, use these groups to search which the collision group is and where the collision occurring, and identify all tags finally.

The remainder of this paper is organized as follows. Section 2 shows the existing anti-collision protocols. Section 3 describes the problems of existing anti-collision protocols and resource. In Section 4, the proposed anti-collision protocol algorithm based on divide and conquer for RFID system is presented. Section 5 presents the experiment results. Finally the conclusions of our analysis are presented in Section 6.

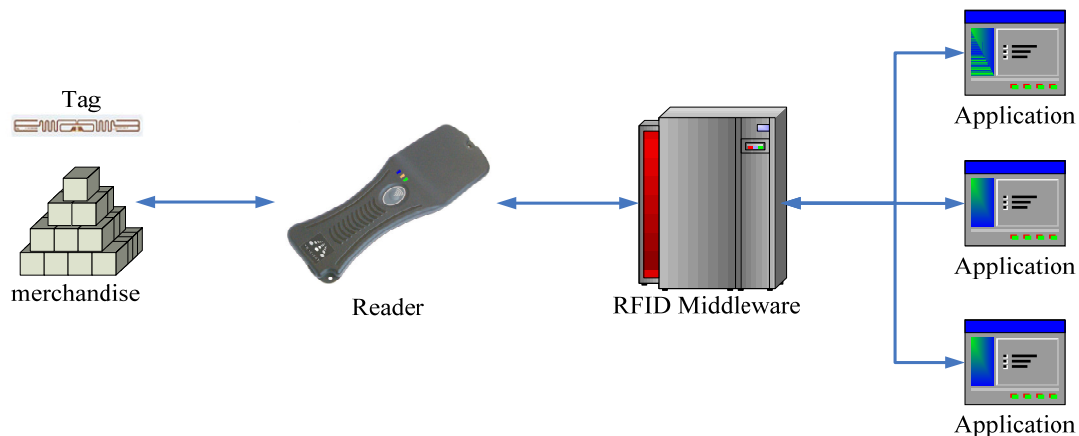


Figure 1 The Parts of RFID System [4][5].

2 Anti-Collision of RFID

In the RFID system, a tag anti-collision protocol for passive RFID tags is used to solve the problem of the order of reading tags. The

responses of multiple tags are often overlapped, such that the reader is unable to read the right tag IDs from the overlapped responding signals. There are three types of tag anti-collision protocols according to the algorithm structure: (1) tag anti-

collision protocols based on binary tree, (2) tag anti-collision protocols based on query tree, (3) tag anti-collision protocols based on probability. These three tag anti-collision protocol algorithms have been used in the RFID systems now.

2.1 Binary Tree Algorithm[11][15]

In the tag anti-collision protocols algorithm based on binary tree, the tags IDs are represented into a tree structure as shown in Figure 2. The reader can read the tags IDs one by one from the response signals of tags, while the tree search algorithm has been modified. While these neighbor multiple tags transmitted their response signals to a reader at the same time, then the response signals of tags are going to be accumulated together as shown in Figure 3. The accumulated signal should let the reader cannot correctly read the response signals and produces the tag collisions. The tag collisions of tree structure always appear at the nodes of a tree as shown in Figure 4; If there are more than one logical voltage levels among the same depth nodes, then the tag collision must exist. Binary tree algorithm resolves the tag collision by eliminating collided bits one by one in a tag ID[15]. When a collision appears, the reader knows the position in which the collision occurs. The tags whose first collided bit is logical 1, they do not respond to the reader's next request. On the other hand, the tags whose first collided bit is logical 0, they transmit their ID to the reader's next request. Repeating this procedure till all the collision bits are solved, then the reader can identify all the tags finally.

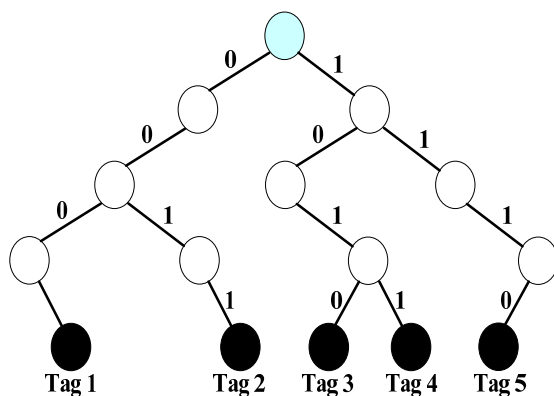


Figure 2 Binary Tree of Tag IDs.

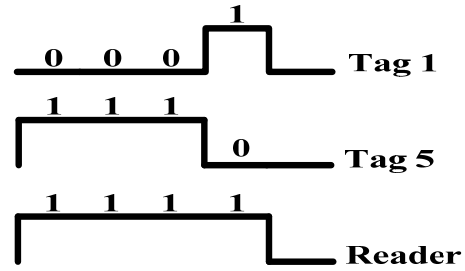


Figure 3 Tags response signals and the reader received signal.

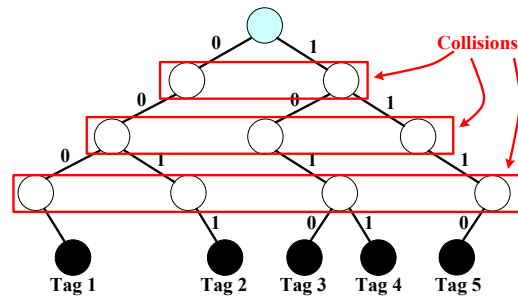


Figure 4. Collisions of IDs.

2.2 Query Tree Algorithm[10][16]

The anti-collision protocols based on query tree algorithm has an advantage in implementation due to their operation simplicity. The query tree algorithm takes the tag IDs as character strings. The query tree algorithm limits a query from the reader and a response from the tag together as one round, and iterates the rounds to identify the tag. In each round, the reader requests the tags whether they contain the certain character string. The tags that have the certain character string should respond to the reader. If more than one tags respond (this is tag collision in the query tree algorithm), the reader recognize that there are more than one tags with the certain character string. Then the reader expands the certain character string with adding '0' character or '1' character to the certain character string, and queries this longer character string to the tags again. If there is exact one tag responds then the reader identifies the tag. The reader expands the character string with adding additional bit to the character string until exact one tag response, and it iterates this process until all the tags within the area are identified. Table 1 shows an example of the corresponding steps of query tree algorithm to identify 5 tags with ID = {01000, 01001, 01011, 01110, 01111}.

In some anti-collision protocols based on query tree algorithm cases, one can use a shorter character strings to finish the tag IDs identification. For the above example, we let the reader to build up the character string by starting from the most significant

bit (MSB), and expanding the character string by adding additional bit to be the least significant bit (LSB) of the character string. Now, we reverse the above procedure. We let the reader to build up the character string by starting from the least significant bit (LSB), and expanding the character string by adding additional bit to be the most significant bit (MSB) of the character string. Table 1 shows the corresponding steps of query tree algorithm to identify the same 5 tags with reversing the above procedure. By comparing Table 1 and Table 2, To identify the same 5 tags, the algorithm with building up the character string by starting from the MSB needs 14 steps, and the algorithm with building up the character string by starting from the LSB only needs 8 steps.

Table 1. Queries and responses between the reader and the tags by searching from the MSB of IDs .

ID = {01000, 01001, 01011, 01110, 01111}		
Step	Query	Response
1	0	collision
2	1	no response
3	00	no response
4	01	collision
5	010	collision
6	011	collision
7	0100	collision
8	0101	01011
9	01000	01000
10	01001	01001
11	0110	no response
12	0111	collision
13	01110	01110
14	01111	01111

Table 2. Queries and responses between the reader and the tags by searching from the LSB of IDs.

ID = {01000, 01001, 01011, 01110, 01111}		
Step	Query	Response
1	0	collision
2	1	collision
3	00	01000
4	01	01110
5	10	01001
6	11	collision
7	110	01011
8	111	01111

2.3 Probabilistic Algorithm[11]

The anti- collision protocols based on probabilistic algorithm is developed from the ALOHA algorithm. The probabilistic algorithm

assumes that the identification probabilities of all tags are the same and the identification probability distribution is a uniform distribution. All the tags of the probabilistic algorithm are also arbitrators. This is the special characteristic of probabilistic algorithm that differed from the above two algorithms. Each collision tag shall decide the next response by itself automatically, and this function of tag is obtained by installing a counter for each tag at the tag implementation procedure. According to the current value in tag's counter, each tag can automatically decide whether to respond reader's request or not. Figure 5 shows an example of the corresponding steps of probabilistic algorithm to identify 5 arbitration tags.

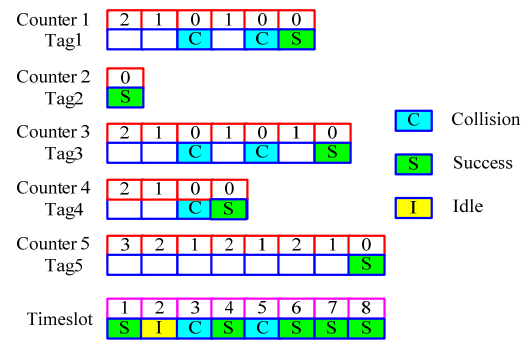


Figure 5. Arbitration Tag IDs by Probabilistic Algorithm.

3 Problems and Resources

All the binary tree anti- collision protocols, the query tree anti- collision protocols, and the probability anti- collision protocols can identify all tags correctly. Nowadays, many papers about the improvement of the identification speed and the collision decreasing have been proposed. Generally, the number of communications and the collisions in a RFID system is increased when the number of tags that need to be identified increase. The complexity and the time consuming of the three kinds anti- collision protocols are unlimited increasing when the number of tags is very large. In this paper, we propose a new anti- collision protocol that different from the above three kinds to decrease the tag collision in RFID system.

We find that the number of tag collision in RFID system should have an upper bound. The upper bound of the number of tag collision is the number of communications, and a communication is the cycle that a reader emits a request and then tags response the request. Collisions should occur at each communication, and the worst case is that each communication has at least one collision. Due to the fact that collisions only occur at

communication procedure, so we are able to decrease the upper bound of the number of collision efficiently if we can decrease the number of communication in the RFID system. And the number of collision must be decrease really while the upper bound of the number of collision. So, to decrease the number of communication in the RFID system for anti- collision is a simpler way than those three kinds of anti- collision protocols.

We have to measure the gain and loss with the consideration of resources of RFID system, to decide that whether the reader really need use its resources to resolve the tag collision problem or not. The result signal after a collision is the accumulation of all the response signals of all the collision tags. The reader receives the accumulation of all the response signals of all the collision tags and is unable to identify any ID from the result signal. The result signal is like as an encrypted message, so that we need other auxiliary tools to decode the encrypted message to find all IDs. Does a reader need to take all the time consuming of the operation of the encrypted message decoding? It is a problem that is worth to be studied by us.

4 Anti-Collision protocol based on Divide and Conquer

With the discussion of resources of RFID system in the last section, we propose an anti-collision protocol algorithm based on search tree. The search tree anti- collision protocol algorithm can effectively reduce the number of communications and the number of collisions in RFID systems.

A RFID starts at the reader to emit a reading signal to tags, tags should response their IDs to the reader after tags receive the reading signal, and then the reader should receive a character string as shown in Figure 6. The character string is composed with character "0", "1", and "C". Where "0" represents the logical 1, "1" represents the logical 0, and "C" indicates that has a collision at that bit. In Figure 6, there are collisions at several bits due to the overlapping of tags response signals, such that the reader is unable to identify the ID packet. In this example, we also find that each collision bit is isolative (all of its adjacent bits are not collision bits). There are not any difficult for the system to identify the ID packet; it only needs to replace the collision bits with logical 0 or logical 1, then it should show all the possible IDs and identify each of them.

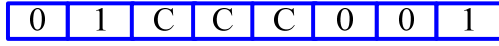
0	1	C	1	C	0	0	1
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C: collision

Figure 6. Tag collision with isolative collision bits.

Figure 6 is an example of tag collision with isolative collision bits. Now, we consider another tag collision example with having adjacent collision bits shown in Figure 6. In Figure 7, there are three collision bits (bit 3, bit 4, and bit 5) and they are successive. A traditional anti- collision protocol algorithm always needs to list all the 8 possible IDs ($2 \times 2 \times 2 = 8$) as shown in Figure 8, and then uses the search tree method to search each of the 8 possible IDs to identify all IDs. Although this algorithm can reduce the number of communications and the number of collisions, but it still waste much resource for some cases of successive collision bits; Figure 9 shows an example that two tags are colliding with three successive collision bits, it is a resource wasting that repeats research procedure eight times to identify two tags.

We point at the resource wasting problem caused by the excessive communication and propose a new anti- collision protocol algorithm based on divide and conquer for RFID system. The divide and conquer algorithm divides a big problem of size n uniformly into two small sub-problems of size k and $n-k$ that can be solved easily and quickly, repeating to the same solving steps to solve these small sub- problems, and then somehow combine their solutions to obtain a solution to the original problem. The reader of the proposed anti- collision protocol algorithm decomposes the collision bits of the received ID into several groups first. Each group is two bit group as shown in Figure 10; the whole groups are 00, 01, 10, and 11. Then, the reader search tags one by one to find which groups are the collision groups and where the collision occurring now. In the research step, the reader sends the possible collision groups and the collision positions to tags at the same time. When there is tags whose IDs match with the reader sent collision groups and collision positions, then those tags just do a response to the reader and don't care the response contents. The response contents are not important for the reader, because the reader only wants to know which the collision group is and where the collision occurring. The reader can get more information from the search and the responses of tags to compose the complete tag IDs and identify them finally. Figure 11 shows an example of tag collision resolved with our proposed algorithm.



C: collision

Figure 7. The response ID with three Successive Collision bits.

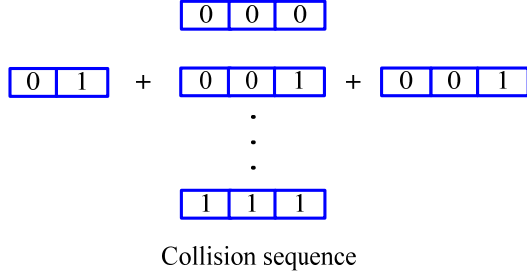
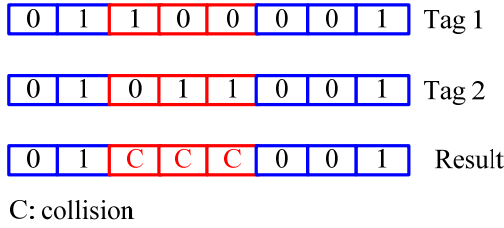


Figure 8. The eight possible IDs for the three Successive Collision bits.



C: collision

Figure 9. An example of two tags with three Successive Collision bits.



C: collision

Figure 10. An example of Collisions with four Successive Collision bits.

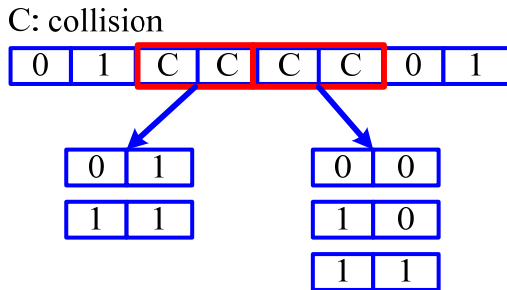


Figure 11. The Possible results of Groups of Collisions for the four Successive Collision bits.

5 Experiment Results

Some standard tags (tag length is 96 bits) are used in simulation for demonstrating the performance of the proposed algorithm. We randomly take $2^k, k = 5, 6, \dots, 10$ standard tags for anti-collision simulation, and take the simulation results to compare with that of query tree algorithm (QT) and that of query tree reversed algorithm (QTR). The times of querying are the accumulative times of the reader querying while all tags are completely identified. The experiment results are arranged in Table 3 and Figure 12. From Table 3 and Figure 12, one can easily find that the querying times of QT and QTR are direct proportion to the number of tags. One also can find that the querying times of DCQT is independent of the number of tags and is a constant number.

Table 3. Comparison of querying times of different algorithms.

	Number of tags					
	32	64	128	256	512	1024
QT	187	249	375	629	1139	2161
QTR	65	129	257	513	1025	2049
DCQT	193	193	193	193	193	193

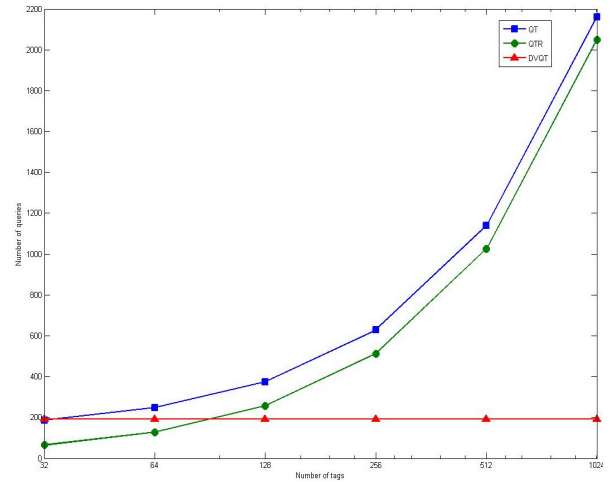


Figure 12. Comparison of querying times among QT, QTR, and DCQT.

6 Conclusion

Our proposed algorithm divides the successive collision bits into several groups, and each group is two bits. The worst environment for a reader is that there are 2^B in its emitting & response receiving area and collision occur at every bit. In the worst environment, there are B groups and each group has 4 possible different contents, so that the number of communications is $2B = B \cdot 4/2$. The upper bound of the number of collisions is the

number of communications, So that the number of tag collision is really no more than the two times of the tag length. The length of a standard RFID tag ID is 96 bits now, and to increase the length of the standard RFID tag ID is possible in the future. The worst condition for a reader now is that there are 2^{96} IDs in its emitting & response receiving area, and the number of tag collision is really no more than $2 \times 96 = 192$ in our proposed algorithm according to the above analysis. In our algorithm, the number of tag collision depends on the length of tags, and is independent of the number of tags. So, the time complexity our algorithm is $\text{BigO}(1)$.

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